MILITARY SPECIFICATION SHEET

ELECTRON TUBE, THYRATRON

TYPE 7590

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the electron tube described herein shall consist of this document and the latest issue of Specification MIL-E-1.

DESCRIPTION: Triode, hydrogen

See figure 1

Mounting position: Any

Weight: 2.25-pounds nominal

ABSOLUTE RATINGS: For crowbar service

Parameter:	Ef	(D) Ebb	epx	epy2	egy	egx	Ecc	ib	Ib
Unit:	Vac	kVdc	kv	kv	v	v	Vdc	a	Adc
Maximum:	6.6	25.0	15.0	30.0	Note 2	450	---	1,000 (Note 3)	0.5
Minimum:	6.0	10.0	---	Note 1	---	---	---	---	---
TEST CONDITIONS:	6.3	25.0	---	---	---	---	0	---	---

ABSOLUTE RATINGS:

Parameter:	tp	tad	tav	tk	TA	dik/dt	Eres	Ires
Unit:	sec (Note 3)	µs	sec	sec	°C	a/µs	Vac	Aac
Maximum:	0.1	1.0	10	---	+75	2,500	5.5	6.5
Minimum:	---	---	---	180	-55	---	2.5	1.5
TEST CONDITIONS:	---	---	---	180	---	---	Note 4	---

GENERAL:

Qualification - Required

(D) denotes changes

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METHOD	REQUIREMENT OR TEST	NOTES	CONDITIONS	AQL (PERCENT DEFECTIVE)	INSPECTION LEVEL OR CODE	SYMBOL	LIMITS MIN	LIMITS MAX	UNIT
	Quality conformance inspection, part 1								
3241	Heater current (cathode)	-		0.65	II	If	12.0	22.0	Aac
3241	Heater current (reservoir)	-	Eres = 4.5 Vac	0.65	II	Ires	2.0	5.0	Aac
(D) ---	DC anode voltage for conduction	5,7		0.65	II	Ebb	---	5.0	kVdc
3268	Energy diverter tubes, gas filled	5,6	t = 30 minutes	0.65	II	egy	---	450	v
---	Reservoir voltage	6	Energy diverter tubes, gas filled	0.65	II	Eres	2.5	5.5	Vac
3251	Pulse emission (method A)	-	ib = 1,000 a; tp = 5.0 µs ±10%; prr = 60 ±10%; tr = 0.5 µs (max); specified time interval = 2.5 µs	0.65	II	egk	---	300	v
	Quality conformance inspection, part 2								
3268	Energy diverter anode delay time	-	Energy diverter tubes, gas filled	---	---	tad	---	2.0	µs
	Quality conformance inspection, part 3								
(D) ---	Life test	5	Group D; t = 96 hours "on" and 1 hour "off" (tube mounted horizontally); t = 1,000 hours	---	---	---	---	---	---
---	Life-test end points:								
3268	Energy diverter tubes, gas filled	-		---	---	egy	---	500	v
---	DC anode voltage for conduction	-		---	---	Ebb	---	10.0	kVdc
3268	Energy diverter anode delay time	-		---	---	tad	---	2.0	µs

NOTES:

1. The maximum forward voltage rating epy2 applies to a transient voltage condition where the duration of the transient does not exceed 2 seconds.
2. The driver pulse, measured at tube socket with thyratron grid disconnected; egy = 550 v (min), 2,500 (max), rate of rise = 1,800 v/µs, tp = 2.0 µs (min), impedance of driver circuit 50 to 200 ohms.
3. The current ratings vary according to the duration of the discharge as listed below:

		ib max
Filter discharge period	0 to 1.5 ms	1,000 a (0.8 coulombs)
Rectifier short-circuit period	1.5 to 100 ms	25 a
Rectifier short-circuit period	1.5 to 50 ms	50 a
Rectifier short-circuit period	1.5 to 30 ms	85 a

Time will be measured from the initiation of the discharge.

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NOTES: -Continued

4. Adjust reservoir voltage to value indicated on tube within ±5 percent.

5. The circuit constants shall be chosen so that at Ebb = 25 kVdc; ib = 500 a (min); dik/dt = 2,500 a/µs. The current shall decay exponentially to 36.8 percent of its peak value in 200 µs. Grid pulse characteristics shall be tp = 2.0 µs (max); tr = 0.25 µs (min); drive impedance = 250 ohms (min).

(D) 6. The optimum reservoir voltage shall be permanently inscribed on the base of each tube. The optimum reservoir voltage shall be within the specified limits. The energy diverter operation shall be conducted at the optimum reservoir voltage.

(D) 7. The dc anode voltage test shall be performed in the same circuit as the operation test for energy diverter tubes and within 60 seconds following the conclusion of the operation test. Under the conditions specified, the anode voltage shall be increased until the tube starts to conduct current when triggered. The dc anode voltage required shall be within the limits specified on the tube specification sheet.

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Navy - EC
Air Force - 85

Review activities:
Air Force - 17, 80
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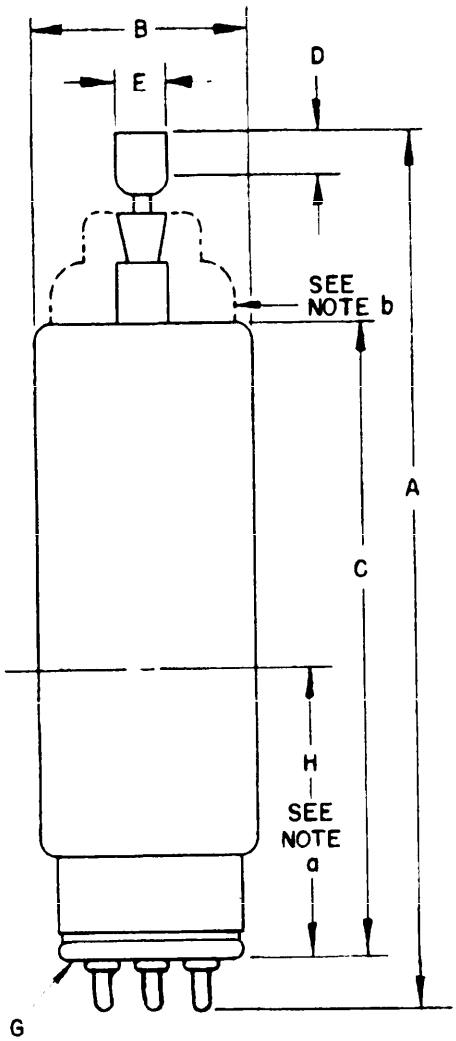
User activities:
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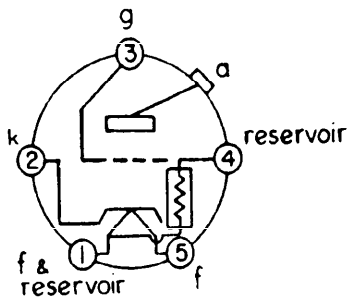
Preparing activity: Navy - EC

Agent: DSA - ES

(Project 5960-2814)

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PIN CONNECTIONS

Ltr	Dimensions in inches with metric equivalents (mm) in parentheses	
	Minimum	Maximum
Quality conformance inspection, part 2		
A	11.50 (292.1)	12.50 (317.5)
B	----	3.313 (84.15)
C	8.00 (203.2)	9.00 (228.6)
(D) E	.559 (14.20)	.573 (14.55)
(D) G	Base A5-98 (EIA)	
H	----	5.00 (127.0)
	Reference dimension	
D	50 (12.7)	

NOTES:
a. Dimension H locates centerline of clamping area. Clamping shall be confined to area 1.50 inches (38.10 mm) above and below this line.
b. Alternate configuration.

(D) FIGURE 1. Outline drawing of electron tube type 7590.

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